Siddhi Sanjay More

Mechanical Systems · Hardware and Mechatronics · Robotics

more.sidd@northeastern.edu | +1 (857) 867-5052 | Boston, MA | Portfolio | LinkedIn | GitHub

Summary: Robotics Engineer with experience in mechanical design, embedded systems, and control systems, including sensor fusion, motor control, CAD/FEA analysis, and ROS2. Experience organizing robotics and physical AI community events, including sponsorship coordination and hardware logistics.

PROJECTS

Smart Ankle Prosthetics, *Northeastern University* **Jan 2026 - Present**

- Built a Simulink/Stateflow gait phase estimator (evolved from an initial MATLAB prototype) classifying heel-strike, mid-stance, toe-off, and swing from fused IMU/FSR data; predicts next phase 75 ms ahead to pre-position the ankle
- Synchronized IMU (MPU6050) and FSR sensors at 200 Hz with zero drift by scheduling concurrent sampling tasks in FreeRTOS on an ESP32-S2; logged timestamped data acquisition (DAQ) streams to CSV via Python
- Architected CAN-FD communication to a Moteus C1 motor controller, transmitting phase suggestions rather than direct commands to keep adaptive torque response onboard

Stanford Pupper v3 x Boston Robot Hackers, *Boston* **May 2026 – Present**

- Developing a Stanford Pupper V3 quadruped for community education; executing hardware assembly, implementing gait control algorithms, and validating kinematics via URDF-based MuJoCo simulations

Sound Triangulation & Doppler Detection, *Northeastern University* **Sep 2025 - Dec 2025**

- Implemented GCC-PHAT TDOA localization with a microphone array to classify hazard sounds and resolve vehicle approach direction within ±10° at under 100 ms latency

Semi-Autonomous Aquatic Cleanup Robot, *Smart India Hackathon* **Jul 2023 - Oct 2023**

- Led design and modeling on a 5-person cross-functional team spanning mechanical, sensing, and navigation subsystems; designed chassis, hull, and crusher mechanism in SolidWorks and validated structural integrity via von Mises FEA before fabrication
- Trained a CNN (TensorFlow/Keras) on 991 images for water hyacinth classification, reaching 87.5% test accuracy

Plastic Injection Molding Optimization via FEA & DOE, *Mumbai University* **Jul 2023 – Jul 2024**

- Reduced sink marks by 20% and warpage by 11% on an automotive airbag housing by running a full 2^3 factorial DOE across melt temperature, mold temperature, and injection pressure in SolidWorks Plastics, then built an ANOVA framework on the results to rank parameter significance and give a clear priority order for process changes without physical trials
- Published as first author, "Experimentation in Injection Molding: Simulation and Parameter Optimization Using Design of Experiment" AIP Conference Proceedings 3397, 040005 (2026)

EXPERIENCE

Revolute (Boston's first robotics + physical AI hackathon), *Co-Organizer*, *Boston* **Jul 2026 – Aug 2026**

- Co-organized a 30-participant robotics hackathon by screening applicants and building the Boston Robot Hackers community partnership through cross-functional collaboration, driving registration for the event
- Independently set up, wired, and tested 2 Jetson Orin Nano + reBot arm stacks, performing hardware-software system integration, and led day-of hardware troubleshooting across the full $15k sponsored fleet alongside the Hardware Lead

IIT Bombay and 1Stop.ai, *Robotics and CAD Intern, Mumbai* **Feb 2022 - Apr 2022**

- Derived DH parameters and built forward/IK solvers for a 4-DOF SCARA robot in MATLAB; validated motion profiles in Robo-Analyzer, achieving sub-mm end-effector accuracy across the full workspace
- Ran static structural analysis in ANSYS Workbench under max-payload load cases; identified a stress concentration that triggered a design revision before fabrication, and designed mechanical parts in SolidWorks against client specifications, applying GD&T and manufacturing tolerancing standards

TECHNICAL SKILLS

Robotics and Software/Programming: Python, C, C++, MATLAB/Simulink, Linux, Git, Arduino IDE, TensorFlow/Keras, ROS/ROS2, Nav2, SLAM, Impedance Control, Gait Phase Estimation, Moteus C1, GCC-PHAT TDOA, Documentation, Microsoft Word

Embedded Systems: ESP32 series, Arduino, MPU6050, FSR, CAN-FD, I2C, L298N, Microcontrollers, RTOS (FreeRTOS), Real-Time Systems, Firmware Development, Hardware Debugging, Hardware bring up, Jetson Orin Nano

Mechanical and Fabrication: Mechanical assemblies, structural analysis, design verification, materials selection, tolerancing, prototyping, laser and foam cutting, FDM (Ultimaker), SLA resin, soldering (SMD and through-hole), PCB milling, hand and power tools

CAD and Analysis: SolidWorks (Motion, Plastics), Fusion 360, CATIA, AutoCAD, NX 11, ANSYS Workbench, Creo, DOE and ANOVA

EDUCATION

Northeastern University **Expected May 2027**

MS in Robotics, Concentration: Mechanical Engineering | GPA: 3.25 | Global Student Award

Courses: Robot Sensing and Navigation, Robotics Mechanics and Control, Wearable Robotics, Control Systems Engineering

Mumbai University, India **Jul 2020 - Jul 2024**

BE in Mechanical Engineering, Honors: Robotics

Courses: FEA, Machine Design, Automation and AI, CAD/CAM, IoT, Supply Chain Management, Product Design, Kinematics of Machinery